CSE430 DSP

Theory Assignment 2

Name: Subhi Bhuiyan

ID: 17201116

Section: 01

Q. Determine the response $y(n)$, $n>=0$, of the system described by the following second order difference equation where the input sequence is $x(n) = \delta(n)$ and the system is initially relaxed.

$$y(n) - 5y(n-2) + 6y(n-3) = x(n) - x(n-2)$$

Ans: The system is initially relaxed i.e. initial conditions = 0

For impulse input, steady state response: $y_s[n] = 0$

Now, for determining zero input solution,

$$\lambda^n - 5\lambda^{n-2} + 6\lambda^{n-3} = 0$$

$$\Rightarrow \lambda^n \left(1 - \frac{5}{\lambda^2} + \frac{6}{\lambda^3}\right) = 0$$

[Multiplying both sides by $\frac{\lambda^3}{\lambda^n}$]

$$\lambda^3 - 5\lambda + 6 = 0$$

$$\lambda = -2.69$$

$$\therefore y_h[n] = c_1(-2.69)^n$$

$$\therefore y[n] = c_1(-2.69)^n \quad \text{——— (i)}$$

Given, $y[n] = x[n] - x[n-2] + 5y[n-2] - 6y[n-3]$ ——— (ii)

From (i), & $y[0] = c_1(-2.69)^0$

$$\Rightarrow y[0] = c_1$$

Scanned with CamScanner

From (ii),

$$y[0] = x[0] - x[-2] + 5y[-2] - 6y[-3]$$

$$= \delta[0] - \cancel{\delta[-2]}^{0} + 5\cancel{y[-2]}^{0} - 6\cancel{y[-3]}^{0}$$

As $\delta[0] = 1$,

$\therefore y[0] = 1$ (Since system is initially relaxed) [$y[k] = 0$ for $k<0$]

Therefore, $C_1 = 1$ ——— (iii)

Again from eqn (i),

$$y[1] = -2.69C_1$$

From eqn (ii), $y[1] = x[1] - x[-1] + 5y[-1] - 6y[-2]$

$$= \delta[1] - \delta[-1] + 5\cancel{y[-1]}^{0} - 6\cancel{y[-2]}^{0}$$

$$= 0$$

Now, $-2.69C_1 = 0$

$\Rightarrow C_1 = 0$ ——— (iv)

When,

$C_1 = 0$	$C_1 = 1$
$y[n] = 0$	$y[n] = (-2.69)^n$

(Ans.)

Scanned with CamScanner